M. Rayhan Zulkarnain

Multi-Platform Software Engineer | Android Mobile, Windows Desktop & Fullstack Web

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Professional Summary

Informatics Management student at Politeknik Negeri Lampung focusing on multi-platform software engineering, spanning Android mobile applications, Windows desktop systems, and the modern web ecosystem. Specializes in architecting comprehensive solutions that bridge intuitive interface design with solid software architecture, from distributed Point of Sale (POS) and production workflow integration, mobile attendance tracking with high data integrity protection (hardware device binding, live visual verification, and precision geolocation validation), to interactive platforms powered by real-time data synchronization.

Technical Arsenal

Mobile & Desktop Systems: Flutter, Dart, Android Native (Kotlin), CameraX API, GPS Geofencing, Electron.js, Thermal Print ESC/POS (Bluetooth & LAN), Offline-First Sync (SQLite 3 / sqflite)

Web & Fullstack Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), Node.js, Express.js, Socket.IO (WebSocket), Tailwind CSS, Three.js / WebGL, Framer Motion, GSAP

Database, Security & Tooling: SQLite 3, Prisma ORM, TiDB Serverless (MySQL), Argon2id, Stateless JWT, Hardware Device Binding, Anti-Cheat Engine, Git, Figma, UI/UX Design

Selected Engineering Projects

SmartPrint POS & Production Management Suite (Desktop & Android Mobile) | *Kasir Rimbun*

Stack: Electron.js, React, TypeScript, Flutter, Dart, SQLite 3, Prisma ORM, Express.js, Socket.IO, Bluetooth/LAN ESC/POS

- Engineered an industrial offline-first Point of Sale (POS) and interactive Kanban production board for a digital printing business, where the Windows desktop app acts simultaneously as the main control station and embedded local LAN server.
- Built the companion Flutter Android cashier & operator app featuring zero-config QR code LAN pairing, two-way real-time Socket.IO sync, automated square-meter (P × L) pricing calculation, background deadline alarms, and Bluetooth thermal receipt printing.

Presensi SMKN 1 Pakuan Ratu | *Anti-Cheat Mobile Attendance & Web Admin Platform*

Stack: Android Native (Kotlin), CameraX, GPS Geolocation, Node.js, TypeScript, Express.js, Prisma ORM, Argon2id, JWT, React.js

- Architected a multi-role mobile attendance and academic scheduling platform with a high-integrity anti-cheat security engine for students, teachers, and school operators.
- Implemented server-side Haversine formula GPS geofencing (100m school radius), mandatory live front-camera selfie capture via CameraX (blocking gallery uploads and mock cameras), and 1-to-1 Hardware Device ID Binding.

COMMBAT (Real-Time Gamified Quiz & Buzzer Platform) | *Provincial Competition Engine*

Stack: React.js, Vite, Tailwind CSS, Node.js, Express.js, Socket.IO, TiDB Serverless, Vercel, Cloudflare Tunnels

- Developed a low-latency, full-duplex WebSocket quiz and digital buzzer system validated live in high-school academic competitions across Lampung Province, with millisecond-precision buzzer arbitration, live leaderboards, and active fullscreen anti-cheat monitoring.

MyKuliah Academic Suite | *Cross-Platform Schedule, Attendance & Task Manager (Android & Windows)*

Stack: Flutter, Dart, Riverpod, Clean Soft Neumorphic 3D UI, Multi-Role Portal (Student & Lecturer)

- Built a cross-platform academic management app featuring automatic active-day routing, live "Now" class indicators, lecturer attendance validation, and assignment deadline countdowns.

Work Experience

Kaesang Cendawan | Software Developer (2023 – Present)

- Developing and maintaining custom e-commerce web applications, ensuring high performance, responsive layouts, and a seamless checkout experience.

Freelance | Fullstack, Mobile & Creative Developer (2025 – Present)

- Architecting cross-platform Android, Windows desktop, and React/Next.js web applications with real-time synchronization and interactive 3D/WebGL visuals.

Rimbun Digital Utama | Graphic & UI Designer (2022 – Present)

- Creating digital assets, brand visual identities, marketing collateral, and production-ready UI layouts for client projects.

Education

Politeknik Negeri Lampung (State Polytechnic of Lampung) | Bandar Lampung, Indonesia

Diploma in Informatics Management (Manajemen Informatika)

Licenses & Certifications

- **International UI/UX Design Competition** | ITechno Cup 2026, Politeknik Negeri Jakarta (Sep 2026)
- **Data - AI Agent for Data Analysis** | IBM SkillsBuild × Hacktiv8 (Oct 2026)
- **Google Certificate (Foundations)** | Google (Jul 2026) • **Dasar Cloud & Gen AI di AWS** | Amazon Web Services (2025)
- **Back-End Pemula dengan JavaScript, Dasar Pemrograman JavaScript & Belajar Dasar AI** | Dicoding Indonesia (Dec 2025)
- **Code Generations & Optimization, Data Science & Analytics, AI for Beginners, Customer Experience (CX) & Business Email** | HP LIFE (2025 – 2026)